Glimpses of Higher Math

PD Resource Sheet

Last updated November 2, 2026

Online, with the interactive Collatz explorer and Goldbach comet: ahitron.github.io/nov26pd

How to use this sheet

Each “glimpse” is a real piece of higher mathematics that fits in a 10-minute warm-up and can stretch to a full period. Each one has a question, a try-it activity, what is known, and ways to adapt it by course.

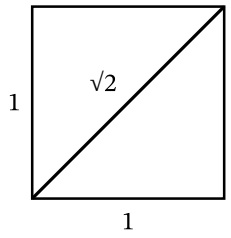
A good glimpse has four features:

- **Low floor.** Any student can start with arithmetic they already know.
- **Real depth.** The question connects to open problems or genuine theorems, not tricks.
- **Pattern, then proof.** Students notice something first, then ask why it must be true (or whether it is).
- **Open questions.** Some of these problems are unsolved, and that’s part of the draw: a sixth grader can understand a question nobody in the world can answer.

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1. The irrationality of $\sqrt{2}$ (Proved)

The claim. No ratio of integers has a square of exactly 2. So the diagonal of a square with sides of length 1, which is $\sqrt{2}$ long, can’t be measured exactly in any unit that also measures the side (incommensurability). The Pythagoreans are usually credited with the discovery, in the 5th century BCE, and Aristotle uses it as his standard example of a proof by contradiction.



By the Pythagorean theorem, the diagonal of a unit square has length $\sqrt{2}$.

The argument.

1. Suppose $\sqrt{2} = a/b$, a ratio of integers in lowest terms.
2. Squaring, $a^2 = 2b^2$, so a^2 is even.
3. So a is even, because the square of an odd number is odd. Write $a = 2c$.
4. Then $4c^2 = 2b^2$, so $b^2 = 2c^2$, and b is even too.
5. Both a and b are even, so a/b was not in lowest terms after all. That contradicts step 1, so no such fraction exists.

Common misconceptions to address

- “Its decimals never repeat, so it’s irrational.” That is a consequence of being irrational, not a proof: no one can check infinitely many digits.
- A calculator’s 1.414213562 is not $\sqrt{2}$. It is a fraction (1,414,213,562 over a billion) whose square is close to 2.
- “Why can we assume lowest terms?” Every fraction can be reduced, so the assumption costs nothing. Ask where the proof uses it.
- This is a genuine proof by contradiction. Compare Euclid’s proof in the next section, which can be done directly.

Try it.

- Hunt for fractions whose square is close to 2, starting from 3/2. Answer key: 3/2, 7/5, 17/12, 41/29. Each time, $a^2 - 2b^2$ is 1 or –1 (9 – 8, 49 – 50, 289 – 288, 1,681 – 1,682): as close as whole numbers can get, but never exact.
- Count the factors of 2. Whatever a is, a^2 has an even number of 2s in its prime factorization, and $2b^2$ has an odd number. So they can never be equal. (A second proof, and a good one for middle school.)
- Last digits: a square ends in 0, 1, 4, 5, 6, or 9. What can $2b^2$ end in? What does that force? ($2b^2$ ends in 0, 2, or 8, so a^2 must end in 0; pushing on shows a and b both end in 0, a third proof.)
- Run the same proof for $\sqrt{3}$. Then try it for $\sqrt{4}$: where exactly does it break?

Extensions

- Better and better fractions: from a/b, the next is $(a + 2b)/(a + b)$, giving 3/2, 7/5, 17/12, 41/29, 99/70, … These are the best approximations to $\sqrt{2}$ for their size.
- A proof without algebra: Tennenbaum’s overlapping squares (https://divisbyzero.com/2009/10/06/tennenbaums-proof-of-the-irrationality-of-the-square-root-of-2/). Put two squares of side b in opposite corners of a square of side a. They overlap in exactly the area they leave uncovered, which gives a smaller pair of whole numbers with the same property, and so on forever.
- A-series paper (A4, the international cousin of Letter) has sides in the ratio $\sqrt{2}$, so folding a sheet in half gives the same shape.
- An open question to end on: nobody knows whether $\pi + e$ is rational.

Adapting it

Course	Angle
Middle school / Algebra 1	Near-miss fractions on a calculator; counting factors of 2; squares and their last digits.
Geometry	The diagonal of the unit square; why A-series paper keeps its shape; Tennenbaum’s picture proof.
Algebra 2 / Precalculus	The structure of a proof by contradiction; $\sqrt{3}$, and why the proof fails for $\sqrt{4}$; the $(a + 2b)/(a + b)$ sequence.

Course	Angle
Computer science	Generate the near-miss fractions and check that $a^2 - 2b^2$ alternates between 1 and −1.

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2. Euclid's proof that there are infinitely many primes (Proved)

The claim. No finite list of primes can contain all the primes. This is Proposition 20 of Book IX of Euclid's Elements, more than 2,000 years old, and still a model of a short, complete proof. Euclid did not say "infinitely many." He wrote that primes are more than any assigned multitude, and argued for three primes standing in for any finite list.

The argument.

1. Take any finite list of primes, say $p_1, p_2, \ldots, p_n$.
2. Form $N = p_1 \times p_2 \times \ldots \times p_n + 1$.
3. N is bigger than 1, so it has at least one prime factor q. (This uses a lemma Euclid proves separately, Book VII, Proposition 31: every number greater than 1 has a prime factor. A quick reason: the smallest divisor of N greater than 1 must be prime, since otherwise it would have a smaller divisor.)
4. If q were on the list, it would divide the product, and also divide N, so it would divide the difference, which is 1. That is impossible.
5. So q is a prime not on the list. Every finite list is incomplete.

Common misconceptions to address

- N does not have to be prime. With the first six primes, 2 × 3 × 5 × 7 × 11 × 13 + 1 = 30,031 = 59 × 509. The proof only needs N to have a prime factor outside the list.
- The proof does not tell you how to find the next prime, or say that multiplying primes and adding 1 generates them. It shows only that a new prime always exists.
- It is often presented as proof by contradiction, but it works directly: given any finite list, it produces a prime not on it.

Try it.

- Give each group a different starting list (for example {2, 5, 11} or {3, 5, 7}). Compute the product plus 1, factor it, and name a prime not on the list. Answer key: 2 × 5 × 11 + 1 = 111 = 3 × 37, which gives two new primes at once; 3 × 5 × 7 + 1 = 106 = 2 × 53. The proof promises a prime not on the list, not the next prime.
- Try {2, 3, 5, 7, 11}: 2,311 is prime. Does that pattern continue? (See Extensions.)
- Ask: where in the proof did we use that the list was finite? What breaks for an infinite list?

Extensions

- Arbitrarily long gaps: for any n, the numbers $n! + 2$, $n! + 3$, ..., $n! + n$ are all composite. There are infinitely many primes, yet they can have huge empty stretches.

- Euclid numbers: take the product of the first n primes and add 1. For n = 1 to 5 you get 3, 7, 31, 211, 2,311, all prime, but n = 6 gives 30,031 = 59 × 509. Whether infinitely many of these numbers are prime, or infinitely many are composite, is an open question.
- Other proofs: Euler showed that the sum of $1/p$ over all primes diverges, and there are other proofs of the infinitude of the primes.
- Compare with the Goldbach and twin prime questions in sections 4 and 5: infinitude of primes is solved, but their finer structure is not.

Adapting it

Course	Angle
Middle school / Algebra 1	Factor products-plus-1 by hand; discuss “what is a proof?” vs. checking examples.
Algebra 2 / Precalculus	Divisibility argument in step 4; the $n! + k$ composite-gap argument.
Computer science	Write a program that takes a list of primes and returns a prime not on it; find where brute force gets slow.

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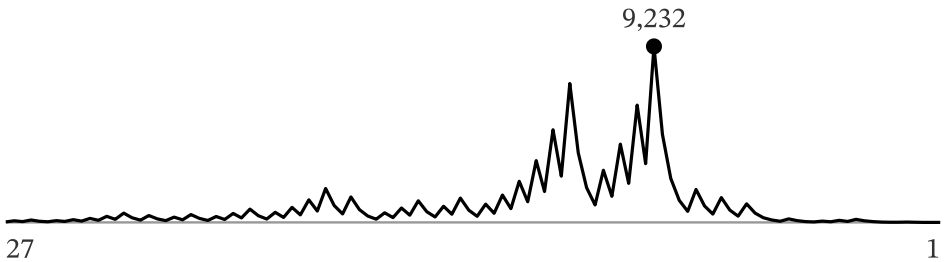
3. The Collatz conjecture (Open problem)

The question. Pick any positive integer. If it is even, halve it. If it is odd, triple it and add 1. Repeat. Does every starting number eventually reach 1?

Nobody knows. Computers have checked an enormous range with no counterexample, but there is no proof.

Try it.

- Start at 6: 6, 3, 10, 5, 16, 8, 4, 2, 1 (8 steps).
- Start at 7: it takes 16 steps and climbs to 52 before falling.
- Start at 27: it takes 111 steps and climbs to 9,232 before falling.
- Under 100, which start takes the longest? (97, with 118 steps.)



The path from 27: 111 steps, up to 9,232, then down to 1. **Open the Collatz explorer** to see the steps for every start, the whole sequence for any number, and what changes with the rules $3n - 1$ and $5n + 1$.

Discussion questions

- Why does 1 lead to the loop 4, 2, 1, 4, …? Could any other loop exist?
- Can the sequence grow forever? What would that require?
- Which starting values are “boring”? (Powers of 2 collapse straight to 1.)
- Change the rule to $3n - 1$. What happens? (There are other loops, for example 5, 14, 7, 20, 10, 5.) What does that say about how special $3n + 1$ is?

What is known. There are four rungs of “known,” and none of them proves the conjecture.

1. **Computer checks.** Every starting value up to 2^{71} (about 2.4×10^{21}) reaches 1. Checking this far is evidence, not proof: Pólya’s conjecture (that up to any point, at least half of the numbers have an odd number of prime factors, counted with repetition) held for every number below 906,150,257 and then failed.
2. **No small loops.** The only known loop is 4, 2, 1. Hercher proved in 2023 that no other loop can have 91 or fewer “up-then-down” blocks, and any such loop would need at least about 7×10^{10} odd steps. So a counterexample would have to be a very long loop or a sequence that grows forever, and nobody can rule out either.
3. **“Almost all” results.** Terras (1976) showed that almost all numbers eventually drop below their starting value. Terence Tao (2019) showed that, in a precise statistical sense (logarithmic density), almost all starting numbers eventually drop below any function that goes to infinity, however slowly. The set of possible exceptions is thin, but it is not shown to be empty.
4. **A heuristic.** Odd n goes to $(3n + 1)/2$ and even n goes to $n/2$, so each step multiplies by about 3/2 or 1/2 with roughly equal odds. The geometric mean is $\sqrt{(3/4)} \approx 0.87$, so sequences behave like random walks that drift downward. Compare $5n + 1$, where the factor is $\sqrt{(5/4)} \approx 1.12$: the drift is upward, and some sequences appear to run off to infinity.

Ask the room: which rung is most convincing, and which would you accept as “knowing” something in math? What would it take to climb from rung 3 to a proof?

Adapting it

Course	Angle
Middle school / Algebra 1	Arithmetic practice with a mystery. Graph the sequence for a few starts and look at the shape.
Geometry / Algebra 2	Parity and “why does every power of 2 work?”; track how many steps each start takes and look for patterns.
Precalculus / Statistics	Plot steps-to-1 for $n = 1$ to 1000 and discuss the distribution; heuristic argument for why the sequence tends to shrink (half the steps multiply by about 3/2 and half by 1/2, so the typical factor per step is $\sqrt{(3/4)} \approx 0.87$, below 1).
Computer science	Write the function, memoize it, and search for the longest sequence under a million. (837,799, with 524 steps.)

4. Goldbach's conjecture (Open problem)

The question. Can every even number greater than 2 be written as the sum of two primes? (Examples: 10 = 3 + 7 = 5 + 5; 28 = 5 + 23 = 11 + 17. Count each way of writing it (each *partition*) once, with the smaller prime first.) Christian Goldbach raised it in 1742, in letters exchanged with Euler.

Try it.

- Write each even number from 4 to 50 as a sum of two primes. Record how many ways each has.
- Which even numbers have only one way? (Only 4, 6, 8, and 12. No larger even number with just one way is known.) Which have the most? (Up to 50: 48, with 5 ways.)
- Why only even numbers? When can an odd number be a sum of two primes? (Parity forces one of the primes to be 2.)

Look at the comet. If you plot the number of partitions against n for even n up to a few thousand, the points spread into bands that widen as n grows, often called the "Goldbach comet." The interactive comet below is a ready-made version; students can also build their own (see In your classroom). Questions to ask:

- Why do the counts tend to grow with n? (More primes below n means more chances.)
- Why is there a clear upper band? (It is the multiples of 3. If n is a multiple of 3 and p is a prime other than 3, then $n - p$ is not a multiple of 3 either, so no such p is ruled out by divisibility by 3. If $n \equiv 1$ or 2 (mod 3), about half of the primes p are ruled out, so multiples of 3 have roughly twice as many partitions.)

Open the interactive Goldbach comet: a plot of how many ways each even number is a sum of two primes. Hover over or tap a dot to see the sums, and use the color options to show the bands.

In your classroom. Students can build the comet themselves: split the even numbers among groups, or write a short program or spreadsheet formula, pool the counts, and plot count against n. This works well as a multi-day activity or for a Math Club or computer science class.

What is known. As with Collatz, there are four rungs of "known," and none of them proves the conjecture.

1. **Computer checks.** Every even number up to 4×10^{18} is a sum of two primes.
2. **Partial theorems.** Vinogradov showed in 1937 that every sufficiently large odd number is a sum of three primes, and Harald Helfgott completed the proof for all odd numbers greater than 5 in 2013 (the "weak" Goldbach conjecture). Chen showed in 1973 that every sufficiently large even number is a prime plus a number with at most two prime factors.
3. **"Almost all" results.** In the 1930s it was shown that the even numbers that fail, if any, have density zero.
4. **A heuristic.** The expected number of partitions of n grows roughly like $n/(\ln n)^2$, so a counterexample becomes less and less plausible as n grows.

A short proof to try. Goldbach implies weak Goldbach. If every even number from 4 up is a sum of two primes, take an odd $n \geq 7$. Then $n - 3$ is even and at least 4, so $n - 3 = p + q$ for primes p and q, and $n = 3 + p + q$. The weak version asks for less (three primes leave more room than two), which is why it could be proved while the original remains open.

Adapting it

Course	Angle
Middle school / Algebra 1	Build a table of Goldbach pairs for small even numbers; discuss why many examples are not a proof.
Geometry / Algebra 2	Parity arguments (why an odd sum of two primes needs a 2); the mod 3 argument for why multiples of 3 have more partitions.
Precalculus / Statistics	Plot partition counts against n and look for the bands; the prime-density heuristic; prove that Goldbach implies weak Goldbach.
Computer science	Verify Goldbach up to a few million; build and plot the comet.

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5. The twin prime conjecture (Open problem)

Another simple-to-state question about primes that remains open.

The question. Twin primes are primes that differ by 2, such as 3 and 5, 11 and 13, 17 and 19. Are there infinitely many pairs?

Try it.

- List the primes below 100 and circle the twin pairs. How many are there? (8 pairs: 3 and 5, 5 and 7, 11 and 13, 17 and 19, 29 and 31, 41 and 43, 59 and 61, 71 and 73.) Do they thin out?
- Look at the gaps between consecutive primes. What gaps appear? Is a gap of 1 possible after 2? Why are all other gaps even?
- Generalize: what about primes that differ by 4 (cousin primes) or 6?

What is known. In 2013 Yitang Zhang proved that infinitely many pairs of primes differ by less than 70 million. Soon after, James Maynard (and, independently, Terence Tao) cut the bound to 600, and the Polymath project refined it to 246. In 2026, new preprints, some found with AI help, have pushed it lower still; none is peer-reviewed yet. Getting the bound down to 2 is the open problem.

Two connections to earlier ideas:

- Euclid extension: $n! + 2$, ..., $n! + n$ shows that gaps between primes can be arbitrarily large. Yet, by Zhang's theorem and its successors, gaps of at most 246 recur infinitely often. That tension is the story.
- Brun proved in 1919 that the sum of the reciprocals of the twin primes converges, while the sum of $1/p$ over all primes diverges (Euler). Twin primes are much sparser than the primes, but a convergent sum does not settle whether there are infinitely many.

Adapting it

Course	Angle
Middle school / Algebra 1	Prime lists and the sieve of Eratosthenes; circle the twin pairs (the sieve at the top of this page works).
Geometry / Algebra 2	Sequences of prime gaps; why every gap after 2 is even; why 3, 5, 7 is the only prime triple with gaps of 2 (of any three odd numbers in a row, one is a multiple of 3).
Precalculus / Statistics	Count twin pairs below 100, 1,000, and 10,000 and compare with the count of all primes; Brun's convergent sum against Euler's divergent one.
Computer science	Sieve implementation; count twin pairs up to a million; find the largest gap between consecutive primes below a million.